Investment Analysis Sub-Note (Ch.2–6, to §6.4)

Scope: Ch.2–Ch.6 (to §6.4), lecture-PPT order. Target: 80-point core. Tags: **DEF** = answer keywords · **CALC** = formula + solving order · **TRAP** = common error · ※ = SA/MC bank item not in PPT (do not use as model answer).

Ch.2 Asset Classes & Financial Instruments

2.1 Money Market

DEF

- Short-term (≤ 1 yr), highly marketable, liquid, low-risk debt = "cash equivalents"
- Large denominations → individuals access via **money-market mutual funds (MMF)**
- Dealer-driven, highly competitive market

Instrument	Keywords
T-bill	4/8/13/26/52 wks; sold at discount, face at maturity; most marketable; ~risk-free; exempt from state & local tax
CD	Time deposit; interest + principal at maturity; negotiable CD ≥ $100,000 marketable
Commercial paper (CP)	Unsecured; large sound firms; ≤ 270 days (usually 1–2 mo); bank line of credit; asset-backed CP = receivables collateral
Bankers' acceptance (BA)	Order to bank to pay at future date; international trade; once accepted = bank liability; trades at discount
Eurodollars	USD deposits at foreign banks / foreign branches of US banks; outside US jurisdiction → less regulation, higher yield; less liquid
Repo / reverse repo	Repo = sell securities + agree to buy back next day at higher price = short-term collateralized (overnight) loan; reverse repo = you lend cash, hold collateral
Fed funds	Reserve balances at Fed; overnight rate between banks; benchmark / monetary policy tool
Broker's call loans	Margin buyers borrow from broker; broker borrows at call money rate; callable on demand

- MMF: stable $1.00 NAV, check-writing, "parking place" for cash
- **TRAP** 2008 "break the buck": Reserve Primary Fund held Lehman CP → NAV < $1 → run → Treasury guarantee ⇒ low-risk ≠ risk-free
- ※ LIBOR (excluded from exam)

CALC — yield conventions (F = face, P = price, n = days)

Yield	Formula	Basis
BDY	$r_{BD} = (F - P)/F \times 360/n$	face value, 360 days; quoting convention only
BEY	$r_{BEY} = (F - P)/P \times 365/n$	purchase price, 365, simple
EAY	$r_{EAY} = (F/P)^{(365/n)} - 1$	purchase price, 365, compound; true annual yield

- Example F = 10,000, P = 9,600, n = 180: BDY = 400/10,000 × 2 = **8.00%**; BEY = 400/9,600 × 365/180 = **8.45%**; EAY = (10,000/9,600)^(365/180) − 1 = **8.63%**
- Order: **BDY < BEY < EAY**
- Discount-quoted (BDY): T-bills, CP, BA. Add-on yield: CDs, repos, fed funds, Eurodollars
- To compare → convert all to BEY or EAY

2.2 Bond Market (fixed-income capital market)

DEF — longer-term borrowing; fixed (or formula) income; segmented by issuer & tax treatment

Type	Keywords
Treasury notes / bonds	Notes ≤ 10 yr, bonds > 10 yr; semiannual coupons; par $1,000; quoted in **32nds**; ~default-free, liquid
TIPS	Principal indexed to inflation (CPI); coupon on adjusted principal; guaranteed **real** return; yield spread vs nominal = expected inflation
Federal agency	GSEs: Fannie Mae, Freddie Mac, FHLB; housing & agriculture credit; implicit guarantee; conservatorship 2008
International	Eurobond (currency ≠ issuer's country), Yankee (USD, sold in US by foreign issuer), Samurai (yen, sold in Japan by non-Japanese issuer)
Municipal	State & local; interest federal-tax-exempt; **GO** = full taxing power; **revenue** = project revenues; lower stated yield

Type	Keywords
Corporate	Semiannual coupon; default risk; rated (investment grade vs high yield); spread over Treasuries; secured (collateral) vs unsecured (debenture, lower priority); callable / convertible / puttable
MBS	Claim on mortgage pool; pass-through of P&I; agencies package conforming loans; securitization; subprime MBS = 2008 crisis

CALC

- Treasury quote: 100-08 = 100 + 8/32 = 100.25% × $1,000 = **$1,002.50** (digits after point = 32nds, not cents; "+" = ½ tick)
- WSJ quote 100.0320: 100 + 3/32 + 0.25/32 = 100.1016% = **$1,001.02**; ≠ 100.032%
- WSJ columns: coupon, bid (dealer buys), asked (you buy), chg (32nds), asked YTM
- TIPS: adjusted principal = P × (1 + inflation); coupon = coupon rate × adjusted principal. Ex: 2% coupon, 3% inflation → 1,030 → coupon **$20.60**
- Muni equivalent taxable yield: r_taxable = r_muni / (1 − t). Ex: 4%, t = 35% → 4/0.65 = **6.15%**; muni wins if taxable yield < cutoff

2.3 Equity Securities

Item	Keywords
Common stock	Residual claim (last in line); **limited liability** (max loss = investment); voting → elects board; dividends discretionary
Preferred stock	Debt + equity hybrid; fixed dividend, perpetuity; cumulative; priority over common, below debt; dividends NOT tax-deductible for issuer; callable / convertible
Priority	Debt → preferred → common (residual)
ADR / GDR	Negotiable certificate from depositary bank for foreign shares held by custodian; ratio sets shares per receipt; economic rights; USD-listed (ADR), London / Luxembourg (GDR); sponsored / unsponsored

- Stock quote fields: Last, Chg, Volume, 52-wk range, P/E = price ÷ EPS, Yield = annual dividend ÷ price
- **CALC** ADR price = underlying home price (in USD) × ratio. Ex: SK hynix 1 ADR = 1/10 share

2.4 Market Indexes

DEF — track market direction; benchmark; basis for index funds & derivatives; design: which stocks, how weighted, how averaged

Index	Weight	Examples
Price-weighted	Share price (high price dominates)	DJIA (30 blue chips)
Value-weighted	Market cap (large cap dominates)	S&P 500 (~80% of US equity value)
Equal-weighted	Equal dollars; avg of returns	—

CALC — setup: A $100 → $110 (20M sh), B $25 → $20 (100M sh)

- Price-weighted: (110 + 20)/(100 + 25) − 1 = **+4.0%**
- Value-weighted: cap A 2,000 → 2,200; B 2,500 → 2,000; 4,500 → 4,200 = **−6.7%**
- Equal-weighted: (10% + (−20%))/2 = **−5.0%**
- DJIA divisor: index = Σprice / divisor. $90 + $30, d = 2 → 60. Split 2-for-1: $45 + $30 = 75; naive 75/2 = 37.5 ×; new divisor = 75/60 = **1.25** (real divisor < 1)
- Others: Nasdaq Composite (tech-heavy), NYSE Composite, Wilshire 5000 (broadest), Nikkei, FTSE, DAX, Hang Seng
- Bond indexes: harder (more bonds, stale prices, matrix pricing, maturity drift, frequent rebalancing); Bloomberg, Merrill Lynch, Salomon

2.5 Derivatives

DEF

- Value derived from underlying asset
- **Call** = right to BUY at strike K; **Put** = right to SELL at K; buyer pays **premium**
- **Futures** = obligation to trade at set price on future date; no premium; exchange-traded, standardized; **marked to market** daily; zero-sum

Position	Payoff at expiry	Profit	Break-even
Long call	max(S_T − K, 0)	payoff − premium	K + premium
Long put	max(K − S_T, 0)	payoff − premium	K − premium

Position	Payoff at expiry	Profit	Break-even
Long futures	$S_T - F_0$	same (no premium)	F_0
Short futures	$F_0 - S_T$	same	F_0

	Options	Futures
Nature	Right	Obligation
Up-front	Premium	Margin only
Payoff	Asymmetric (capped loss)	Symmetric (linear)
Adverse move	Let expire	Must settle
Analogy	Insurance	Locked-in price

- **TRAP** Option writer (seller) has the obligation, not the buyer; call loss capped at premium only for the buyer

Ch.3 Securities Markets

3.1 Issuing Securities

	Primary market	Secondary market
Trade	New issues sold to investors	Existing securities among investors
Issuer	Receives proceeds	No new money
Role	Investment bankers assist	Liquidity, price discovery

- **IPO process**: ① file prospectus with SEC → ② underwriters market, **book-building** → ③ offer price set; **firm commitment** (underwriter bears placement risk) → ④ list & trade in secondary market
- **Underpricing**: IPO priced below first-day trading price → large first-day return = cost to issuer, benefit to allocated investors
- Firm commitment: syndicate buys entire issue, resells, spread = pay, underwriter bears unsold risk. Best efforts: agent only, unsold shares return to issuer (※ best efforts not in PPT)
- **CALC** IPO money left on table = (P_first-day close − P_offer) × shares. Ex: 20 → 26, 10M sh → **$60M**; first-day return = 26/20 − 1 = **30%**
- ※ Winner's curse, long-run IPO underperformance: not in PPT

3.2 How Securities Trade

Market type	Keywords
Direct search	Find each other; sporadic, low-value; lowest organization
Brokered	Broker searches for a fee; block trades, real estate
Dealer	Bid/ask quotes from inventory; profit = spread
Auction	All traders converge, one place & price; most integrated

CALC — bid / ask

- You **buy at ask** (higher), **sell at bid** (lower)
- Spread = Ask − Bid = dealer gross profit = investor round-trip cost. Ex: 50.10 − 49.90 = **$0.20**

Electronic trading

- **ECN**: matches orders directly, no dealer
- **Algorithmic**: timing, price, size → cut cost & market impact
- **HFT**: ultra-fast, tiny fleeting price gaps, milliseconds
- Benefits: speed, lower cost, transparency, anonymity for large orders, after-hours trading

Venue	Keywords
NYSE	Auction / order-driven; central limit-order book; **price–time priority**; hybrid (electronic + human oversight at open / close); blue chips
Nasdaq	Dealer / quote-driven; competing **market makers**; no floor; screen-based since 1971; tech / growth
KRX	KOSPI (large cap), KOSDAQ (growth / tech); auction; opening & closing call auctions; also clearing, derivatives

Venue	Keywords
Nextrade (NXT)	Korea's 1st ATS; Mar 2025; 34-firm consortium; 08:00–20:00; ~800 stocks; maker–taker fees; mid-price & stop-limit orders
Smart order routing	Broker auto-routes to better price (KRX vs NXT) → duty of **best execution**
Cboe	1973; 1st standardized options; largest US options exchange; **VIX** = fear gauge, expected 30-day S&P 500 volatility
CME Group	Largest futures exchange; CME + CBOT 2007 (+NYMEX); Globex; 1982 first equity-index futures; clearinghouse = central counterparty; CFTC

- Options / futures: **options** = right, premium, loss capped; **futures** = binding, standardized, marked-to-market, centrally cleared
- Regulator: securities & security options → **SEC**; futures & commodities → **CFTC**

3.3 Order Types

	Market	Limit	Stop
Execution	Immediate, best available price	Buy ≤ / sell ≥ set price	Dormant until trigger
Certain	Execution	Price	—
Uncertain	Price (**slippage**)	Execution (may never fill)	Price (becomes market)
Use	Liquid stock, fill matters	Price-sensitive, patient	Limit loss / protect gain

- **Stop-loss** → market order after trigger; **stop-limit** → limit order after trigger (price guaranteed, fill not)
- **Marketable limit** (crosses spread) fills now; **non-marketable** rests in book, adds liquidity
- Trade-off: market = certainty of fill; limit = certainty of price

CALC

- Fill rule: buys hit lowest asks; sells hit highest bids; average price = Σ(p × q)/Σq
- Slippage: ask book 100@50.10, 200@50.15, 500@50.25; market buy 500 → (5,010 + 10,030 + 10,050)/500 = **$50.18** vs last trade 50.10
- Limit orders at bid 49.90 / ask 50.10: buy @49.95 rests; buy @50.10 fills 50.10; sell @50.05 rests; sell @49.90 fills 49.90
- Book (bids 50.10×500, 50.05×300, 50.00×800; asks 50.20×400, 50.25×600, 50.30×700): market buy 600 → (400 × 50.20 + 200 × 50.25)/600 = **$50.217**; market sell 1,000 → (500 × 50.10 + 300 × 50.05 + 200 × 50.00)/1,000 = **$50.065**; limit buy 500 @50.15 → no fill (ask 50.20 > 50.15), rests as best bid
- Gap-down: own @80, stop 72, opens 68 → stop-loss fills ≈ **68**; stop-limit (72/72) **no fill**
- Scenarios: cap loss = stop-loss; buy only ≤ 45 = limit buy; must own now = market; lock in gain = stop-loss (sell stop)

3.4 Margin Trading

DEF

- Buy with own **equity** + **broker loan**; shares = collateral; pay interest
- **Initial margin** (Fed Reg T) ≥ 50%; **maintenance margin** lower (25–40%); below → **margin call** (add cash / securities or position sold)
- Loan fixed → equity absorbs all gains & losses → **leverage magnifies both**; profitable only if stock return > borrowing rate

CALC

1. Balance sheet: Assets = Loan + Equity
2. Margin % = Equity / Asset value = (V − Loan)/V
3. Buying power = Cash / initial margin = 1 / m_init (50% → 2×, 40% → 2.5×, 33% → 3×, 25% → 4×). Ex: cash 10,000 @50% → buy **20,000**, borrow 10,000
4. 400 sh @50 (loan 10,000): at $45 → equity 18,000 − 10,000 = 8,000; margin = 8,000/18,000 = **44.4%**
5. **Margin-call price (long)**: (N·P − Loan)/(N·P) = m → **P = Loan / [N(1 − m)]**. Ex: 10,000/(400 × 0.7) = **$35.71** (call when price falls below)
6. Return on equity = [V_1 − Loan(1 + r_b)]/Equity_0 − 1. Ex: price ±20% → ±**40%** (ignoring interest)

- **TRAP** Margin denominator = total asset value; call-price formula uses (1 − m)

3.5 Short Selling

DEF

- Borrow shares → sell high → buy back low (**cover**) → return; profit if price **falls**; motive: bearish view or hedge
- Owe shares, not cash; must reimburse dividends (payment in lieu); borrow fee (higher if hard-to-borrow); forced buy-in on recall; no fixed maturity
- Done in margin account; proceeds held as **collateral** (not withdrawn) + initial margin deposit (Reg T 50% of short value)
- Payoff: max gain **limited** (price → 0); loss **unlimited**
- Risks: unlimited loss, margin call on **price rise**, **short squeeze**, forced buy-in, fees & dividends, bans (uptick rules, halts)

CALC (short 1,000 sh @ $100)

1. Collateral = proceeds 100,000 + margin 50,000 = **150,000**
2. Equity = Collateral − Value owed = 150,000 − 1,000·P
3. Margin = Equity / Value owed. Start 50,000/100,000 = 50%; at $110 → 40,000/110,000 = **36.4%**
4. **Margin-call price (short)**: (C − N·P)/(N·P) = m → **P = C / [N(1 + m)]**. Ex m = 30%: 150,000/1,300 = **$115.38** (call when price rises above)
5. Return = (P_sell − P_cover − dividends/share)/(initial margin per share). Ex: 100 → 90 → 10,000/50,000 = **+20%**

- **TRAP** Long: (1 − m) in denominator, call on fall. Short: (1 + m), call on rise

3.6 Regulation

Pillar	Keywords
Securities Acts 1933 & 1934	Full disclosure at issue; created **SEC**
Sarbanes-Oxley 2002	Corporate governance, auditing, financial reporting after scandals
Self-regulation (FINRA)	Industry polices member conduct & trading practices
Insider trading rules	Prohibit trading on **material non-public information**

Ch.4 Mutual Funds & Other Investment Companies

4.1 Investment Companies

DEF

- Financial intermediary: pools funds of many investors → invests in wide range of securities
- **Pro-rata claim**: own 1% of shares = 1% of every asset + 1% of income; no specific securities owned; no guaranteed return

#	Four core services
1	Record keeping & administration (reports, distributions, reinvestment, tax reporting, switching within family)
2	**Diversification & divisibility** (fractional share of many securities)
3	**Professional management** (analysts, PMs) — beating market = empirical question, not guaranteed
4	**Lower transaction costs** (large blocks)

CALC — NAV

- **NAV = (Market value of assets − Liabilities) / Shares outstanding**
- Assets marked to market at close; liabilities = accrued fees, payables, borrowing
- Ex: assets 35.8M, liabilities 0.8M, shares 3.5M → 35.0/3.5 = **$10.00**
- Ex: 412M assets, 37M liabilities, 15M shares → 375/15 = **$25.00**; portfolio +4%, liabilities same → (412 × 1.04 − 37)/15 = 391.48/15 = **$26.10**
- New shares issued at NAV → assets & shares rise proportionally → NAV unchanged, no dilution
- Open-end: transact at NAV (struck once daily at 4:00 p.m. ET); closed-end & ETF: market price

4.2 Types of Investment Companies

Type	Keywords
UIT	Fixed, **unmanaged** portfolio in trust by sponsor; redeemable units; **self-liquidating** (set termination date); uniform assets (e.g., munis); very low fees; sold at NAV + 1–4% markup

Type	Keywords
Open-end (mutual fund)	Shares issued / redeemed; buy from / redeem with fund; price = **NAV** (once daily 4pm); all orders same price; no intraday trading
Closed-end	Fixed shares after IPO; no issuance / redemption; trade on exchange with other investors; price by supply & demand → **premium / discount** to NAV; intraday, limit orders, short sales
Commingled funds	Pooled vehicles for small trust / retirement accounts, managed collectively
REIT	Pools funds in income-producing real estate; diversified exposure without managing property
Hedge fund	Private pool; **accredited / institutional** only; not registered under Investment Company Act; short sales, derivatives, heavy leverage

	Mutual fund	Hedge fund
Investors	General public	Accredited / institutional
Regulation	Investment Company Act 1940, full disclosure	Largely unregulated, limited disclosure
Liquidity	Daily redemption at NAV	**Lock-up**, quarterly redemption with notice
Leverage / shorts	Tightly restricted	Essentially unrestricted
Fees	Expense ratio ~0.03–1.5%	**"2 and 20"**: ~2% of assets + performance fee

4.3 Mutual Funds

Policy	Keywords
Money market	Short-term, high-quality debt; liquidity, capital preservation
Equity	Income funds (dividends) vs growth funds (capital appreciation)
Sector	Single industry; high concentration risk
Bond	By issuer (gov, corp, muni, MBS), maturity, credit quality
International	Global (incl. US), international (ex-US), regional, emerging
Balanced	Stocks + bonds, relatively **stable mix** (※ not in PPT as a named type)
Asset allocation / flexible	Multiple asset classes; adjusts mix strategically / tactically
Index	Tracks benchmark; low turnover, research cost, fees

- Policy (prospectus) drives risk exposure more than manager skill
- Indexed assets overtook active for the first time (ICI): ~$21.8tn indexed vs $18.8tn active (May 2026)
- Sold: **direct-sold** (fund company, no broker, lower distribution cost); **broker-sold** (loads, conflict of interest); **fund supermarkets** (many families, no-transaction-fee)

4.4 Costs of Investing

Fee	Keywords
Expense ratio	Admin + advisory; deducted **daily** from assets, never billed → investors underestimate; reported return already **net**
Front-end load	Commission at purchase; legal cap **8.5%**, rarely > 6%; reduces amount invested
Back-end load (CDSC)	Redemption fee; ~5% declining 1 pt/yr (5, 4, 3, 2, 1, 0); **locks in** investor
12b-1 fee	SEC Rule 12b-1; annual, max **1%** of assets; marketing / adviser pay; inside expense ratio; economically = load paid in instalments
Soft dollars	In roadmap only; no detail in PPT

- Loads go to selling broker, not fund; published (NAV-based) returns **overstate** load-fund investor's return

CALC

- Front-end load: invested = P × (1 − load). 5.75% on 10,000 → **$9,425**; break-even = 10,000/9,425 − 1 = **6.10%** (> load %)
- Offer price = NAV/(1 − load)
- **Net return ≈ Gross − Expense ratio (− 12b-1 per PPT)**; loads applied separately to initial & final cash flows
- **Fund HPR = (NAV_1 − NAV_0 + income dist. + cap-gain dist.)/NAV_0**. Ex: (30.10 − 28.50 + 0.60 + 1.20)/28.50 = **11.93%** (net); gross = 11.93 + 1.20 = **13.13%**; with 5% front load: buy price 28.50/0.95 = 30.00 → 31.90/30.00 − 1 = **6.33%**
- Distributions must be added back (NAV drops by distribution on ex-date)

- **Share class**: A = 4% front + 1.00% exp → net 11.00%; C = no load, 1.75% exp, 1% CDSC yr1 → net 10.25% (12% gross)
  - A_n = 10,000(0.96)(1.11)^n; C_n = 10,000(1.1025)^n (yr1 × 0.99 for CDSC)
  - 1 yr: A 10,656 < C 10,915; 3 yr: 13,129 < 13,401; 6 yr ≈ break-even (17,956 vs 17,959); 10 yr: 27,258 > 26,533; 20 yr: 77,399 > 70,400
  - **Short horizon → Class C; long horizon → Class A** (load one-time; annual fee compounds)

4.5 Exchange-Traded Funds

DEF

- ETF: trades on exchange all day (like closed-end) near NAV (like open-end); **investors transact with each other, not the fund**
- **Creation / redemption in kind** by **Authorized Participant (AP)**: swaps basket of underlying shares ↔ ETF shares, no cash
- **Arbitrage**: ETF > basket → AP buys basket, delivers for new ETF shares, sells ETF → price falls back; ETF < basket → reverse

Gains	Costs
Trade intraday: limit, stop, short, options	Spread + commission on every trade
No cost of others' exits → **tax efficiency** (in-kind, no forced selling → fewer cap-gain distributions)	Arbitrage not instant / free (e.g., Korea-listed US-index ETFs trade while NY closed → no live NAV)
No idle cash buffer (mutual fund holds 1–5%)	Easy trading → overtrading → lower returns
	Label = how it trades, not what it holds (single-stock 2× ETF)

- Concentration: BlackRock, Vanguard, State Street = 59% of global ETF assets
- Sorting ETFs: what it holds (equity / bond), how run (passive / active), how built (plain / covered call / leveraged / inverse)

CALC — Leveraged ETF (2× resets **daily**; ±10% days)

- Index: 100 → 110 → 99 → 108.9 → 98.01 = **−1.99%**
- 2× ETF: 100 → 120 → 96 → 115.2 → 92.16 = **−7.84%**
- Volatility decay: compounding on changing base; multi-period ≠ 2× index return

CALC — Covered call (own stock @100, sell call K = 105, premium 2)

- Payoff = (S_T − 100) + 2 − max(S_T − 105, 0)
- S_T: 80 → −18; 90 → −8; 100 → +2; 105 → +7; 120 → +7; 140 → +7
- Max gain = (K − S_0) + premium = **+7**; **upside capped, downside open**; premium constant = not protection
- Payout = option premium, **not dividend**; best when flat, worst in strong rally

4.6 Investment Performance

Topic	Keywords
Active vs passive	SPIVA: 79% US large-cap active underperformed S&P 500 (2025); 92% of US domestic active underperformed over 20 yrs; worse with longer horizon
Arithmetic of active mgmt	Passive holds market → active as a group holds market → value-weighted avg gross active return = market return; **average active return = R_market − costs**
Limits	Says nothing about any single fund or dispersion; winners paid for by losers
Survivorship bias	Failed / merged funds drop from list (16% within 10 yr, 47% within 15 yr); visible funds are filtered; SPIVA counts them as underperformers
Persistence	Top-quartile 2021 (164 funds): 20.1% still top in 2022, **0.0%** in 2023; chance = 0.25^2 = **6.25%** → past winners don't repeat; fee is the lasting signal
Coin analogy	1,000 coins × 5 flips → $1{,}000/2^5 \approx$ **31** all heads → luck, not skill
Korea	77% of Korea-domiciled US equity funds underperformed S&P 500 (won terms)

- **CALC** Expected number of repeaters by chance = N × 0.25^k (k years top-quartile). Ex: 164 × 6.25% ≈ **10**

Ch.5 Risk, Return & the Historical Record

5.1 Rates of Return

DEF — HPR

- HPR = (P_1 − P_0 + Div)/P_0 = **capital gain yield + dividend yield**
- Denominator = **beginning (purchase) price**, never ending price
- Ex: 70,000 → 76,000, div 1,500 → gain 8.57% + div 2.14% = **10.71%**

Average	Formula	Keywords
Arithmetic	r̄ = Σr_t / n	Simple mean; **forecasts** next-period E(r); overstates compound growth
Geometric (time-weighted)	g = [Π(1 + r_t)]^(1/n) − 1	Realized compound growth; **past performance**; fund disclosure standard; g ≤ r̄, gap grows with volatility
Dollar-weighted (IRR)	PV(inflows) = PV(outflows)	Reflects investor's cash-flow timing; grades **investor**; differs from TW when flows added / withdrawn

- Rough link: **g ≈ r̄ − ½σ²** (volatility drag)
- Time-weighted → judge **manager**; dollar-weighted → judge **investor** (contaminated by client timing)

CALC

- Returns +25, −15, +30, −5: arithmetic = 40/4 = **8.75%**; wealth = 1.25 × 0.85 × 1.30 × 0.95 = 1.3122 → geometric = 1.3122^(1/4) − 1 = **7.03%**; check 8.75 − ½(0.1916)² ≈ 6.9%
- IRR ex: CF0 = −50,000; CF1 = −53,000 + 2,000 = −51,000; CF2 = 108,000 + 4,000 = +112,000 → solve 0 = −50,000 − 51,000/(1 + r) + 112,000/(1 + r)² → **r ≈ 7.1%**; time-weighted: sub-period 1 = (53,000 + 2,000 − 50,000)/50,000 = 10%, sub-period 2 = (2×54,000 + 4,000 − 2×53,000)/106,000 = 5.66% → √(1.10 × 1.0566) − 1 = **7.81%**

APR vs EAR

- APR = simple scaling, ignores compounding; **EAR = (1 + APR/n)^n − 1**
- Continuous: **EAR = e^APR − 1**; APR = ln(1 + EAR)
- EAR rises with n, converges as n → ∞; compare products on EAR; continuously compounded returns **add** over time
- APR 12%: n = 1 → 12.000%; 2 → 12.360%; 4 → 12.551%; 12 → 12.683%; 365 → 12.747%; ∞ → **12.750%**
- Ex: card APR 19.9% monthly → (1 + 0.199/12)^12 − 1 = **21.82%**

5.2 Inflation & Real Rate

DEF — Fisher equation: (1 + R) = (1 + r)(1 + i)

CALC

- Exact: **r = (1 + R)/(1 + i) − 1**; approx: r ≈ R − i
- Approximation **overstates** real return; error = r × i; negligible at low inflation; never for long-horizon / cumulative
- Ex: R 8%, i 3.5% → 4.35% (approx 4.50); R 20%, i 15% → 4.35% (approx 5.00) ⇒ same real return, different nominal; R 3.5%, i 5.1% → **−1.52%**
- Equilibrium nominal rate: **R = r_real + E(i)**; expected inflation passes 1-for-1 into R; real rate set by saving & investment; keeps lenders' real return

5.3 Risk & Risk Premiums

CALC — scenario analysis (list states s, p(s), r(s))

- **E(r) = Σ p·r**
- **σ² = Σ p·[r − E(r)]²**; **σ = √σ²**
- Ex: Boom .25/+40, Normal .35/+15, Slowdown .25/−5, Recession .15/−25 → E = 10 + 5.25 − 1.25 − 3.75 = **10.25%**; σ² = 221.27 + 7.90 + 58.14 + 186.38 = 473.69 → **σ = 21.76%**
- **Excess return** = r − r_f (realized, can be negative); **risk premium** = E(r) − r_f (expected, positive). Ex: 10.25 − 3 = **7.25%**
- Risk averse: needs positive premium, rejects fair game (standard assumption) · risk neutral: E(r) only · risk loving: accepts fair game

Normality

- 68-95-99.7: E ± 1σ / 2σ / 3σ (ex: −11.5–32.0; −33.3–53.8; −55.0–75.5)
- Convenient: 2 parameters define distribution; symmetric σ = full risk; sum of normals is normal
- Ex under N(10.25, 21.76²): P(r < 0) = Φ(−0.47) = **31.9%**; P(r > 32) = z = 1 → **15.9%**; P(r < −33.3) = z = −2 → **2.3%**; 95% CI = 10.25 ± 1.96σ = **−32.4 to 52.9%**

Departure	Formula	Keywords
Skewness	$E[(r - E)^3]/\sigma^3$	Asymmetry; negative = long left tail, mean < median < mode; equities left-skewed → σ understates risk
Kurtosis	$E[(r - E)^4]/\sigma^4 - 3$	Normal = 0; **fat tails**, extremes more frequent → tail risk

- Evidence: Black Monday 1987 (≈ −20.4%, −20σ), 2008 / 2020 KOSPI crashes → normal understates extreme losses

VaR (Value at Risk)

- **DEF**: loss exceeded with probability α over horizon T (α-quantile of loss); needs **horizon T, confidence c, base value V_0**
- 1-day 99% VaR ₩40M → exceeded ~1 day in 100; **not maximum loss**; silent on size beyond cut-off
- **Parametric VaR = $V_0 \times z_c \times \sigma \times \sqrt{T}$** (mean ignored: σ ∝ √T, μ ∝ T)
- z: 95% = **1.645**; 99% = **2.326**; 99.5% = 2.576
- √T rule assumes independence & constant σ (fails in crisis); **TRAP** do not multiply by T
- Ex: V_0 = ₩1bn, σ = 2%/day: 95% 1-day = 1.645 × 0.02 × 1bn = **₩32.9M**; 99% 1-day = **₩46.5M**; 99% 10-day = 46.5 × √10 = **₩147.1M**
- Limits: normality understates tails; no loss size past cut-off
- **Historical simulation**: sort actual past returns, read quantile; no distribution assumed, fat tails free, needs long sample
- **Monte Carlo**: simulate paths from assumed model; handles options / non-linear; model risk
- **Expected shortfall (ES / CVaR)**: average loss beyond VaR; **coherent (subadditive)**; Basel FRTB: 99% VaR → **97.5% ES**
- Others: max drawdown, Sortino, stress tests

Sharpe ratio

- **$S = [E(r_P) - r_f]/\sigma_P$** = reward per unit of volatility = slope of CAL
- Unchanged by mixing risk-free → measures quality of risky portfolio
- Fails: single security (denominator includes diversifiable risk), assumes normality (flatters fat left tails), interval-dependent, estimation error
- Ex: 7.25/21.76 = **0.33**

5.4 Historical Record (US, 1928–2021)

Asset	Avg return (arith.)	σ
Small caps (bottom decile)	18.5%	38.5%
Large stocks (S&P 500)	11.8%	19.5%
Baa corporates	7.2%	7.4%
10-yr T-bonds	5.1%	7.7%
3-mo T-bills	3.4%	3.1%

- Lessons: higher return ⇔ higher risk; **equity risk premium over T-bills = 8.44 pp** (11.82 − 3.38); T-bills 3.38% vs ~3% inflation → real ≈ 0
- S&P 500: arithmetic **11.82%** > geometric **9.98%**; $100 → $761,748
- Baa vs T-bond gap 2.1 pp = credit + term + liquidity premium (σ not comparable)

5.5 Asset Allocation

DEF

- Complete portfolio C: fraction **y** in risky P, **(1 − y)** in risk-free
- Risk-free: σ = 0, Cov = 0 → **$\sigma_C = y \cdot \sigma_P$**; y > 1 = borrowing
- Sharpe ratio **invariant to y** → choose P and y separately = **separation property**

CALC — complete portfolio

- **$E(r_C) = r_f + y[E(r_P) - r_f]$**; **$\sigma_C = y\sigma_P$**; premium_C = y·premium_P
- P: 14%, 20%, r_f 4%, y = 0.7 → E = **11%**, σ = **14%**, premium **7%**, Sharpe **0.50**; wealth 10m → 3.0m r_f + 7.0m P (60/40 → 4.2m equity, 2.8m bond); y = 0.5 → E 9%, σ 10%

CAL (capital allocation line)

- **$E(r) = r_f + S \cdot \sigma$**; intercept = r_f; slope = S = (14 − 4)/20 = **0.50**; left of P lending, right borrowing
- **Kink** at P: borrowing rate > r_f → slope falls beyond P

Utility

- **$U = E(r) - \frac{1}{2} \cdot A \cdot \sigma^2$** (decimals); A > 0 averse (typical 2–4), A = 0 neutral, A < 0 loving
- ½ A σ^2 = risk penalty (grows with σ^2); risk-free U = r_f
- **Certainty equivalent** = risk-free rate giving the same U

- Ex: P3 (13%, 20%), A = 8: 0.13 − ½·8·0.04 = **−3%**; table (U): P1 (7/5): 6.75 / 6.50 / 6.00; P2 (9/10): 8.00 / 7.00 / 5.00; P3 (13/20): 9.00 / 5.00 / −3.00 for A = 2 / 4 / 8 → choose P3 / P2 / P1

Optimal y*

1. U = r_f + y[E(r_P) − r_f] − ½ A y² σ_P²
2. dU/dy = 0 → **y* = [E(r_P) − r_f] / (A·σ_P²)**

- y* ↑ with risk premium; ↓ with A; ↓ with σ²; y* > 1 → borrow
- Ex: 0.10/(A × 0.04): A = 2 → **1.25** (E 16.5%, σ 25%, U 10.25%); A = 4 → **0.625** (E 10.25%, σ 12.5%, U 7.13%); A = 6 → **0.417** (E 8.17%, σ 8.33%, U 6.08%)
- Borrowing rate above r_f → y* capped below 1.25 (kink)

5.6 Passive Strategies & CML

- **Passive** = hold broad market index, no security analysis / timing
- **CML** = CAL when P = broad market index
- Rationale: **low cost** (0.03–0.2% vs 1%+ active); **free-rider** (others' analysis already in price); low turnover; automatic diversification
- Open: **Grossman–Stiglitz paradox** (if all passive, who sets prices?); index may not be well diversified; passive ≠ no risk

Ch.6 Efficient Diversification (§6.1–6.4)

6.1 Diversification & Portfolio Risk

	Systematic (market)	Firm-specific (idiosyncratic)
Source	Economy-wide: business cycle, rates, inflation, FX, oil, war, pandemic	One company / industry: CEO exit, failed drug trial, fire, lawsuit, fraud
Diversifiable?	**No** (non-diversifiable)	**Yes** (good / bad surprises offset)
Priced?	**Earns risk premium**	Not rewarded

- **Var(total) = Var(systematic) + Var(firm-specific)** (uncorrelated parts)
- **Insurance principle**: risk reduction by spreading exposure across many independent risk sources
- Enron 2001: ~60% of 401(k) in Enron stock; $90 → < $1; job + savings on same firm → doubling down
- Prices set by diversified investors → only systematic risk priced

CALC — n equally weighted stocks (w = 1/n)

- **σ²_P = (1/n)·avg variance + (1 − 1/n)·avg covariance**
- n → ∞: first term → 0; floor = **average covariance** (σ_P = √ avg Cov) = market risk
- Ex: σ = 40% (var 0.1600), ρ = 0.30 → avg Cov = 0.30 × 0.4 × 0.4 = **0.0480**
  - n = 1: 40.00%; n = 10: 0.0592 → **24.33%**; n = 50: 0.0502 → 22.41%; n = 100: 0.0491 → 22.16%; n = ∞: 0.0480 → **21.91%** (floor)
  - 1 → 10 stocks removes most diversifiable risk; 100 → ∞ almost nothing

6.2 Two Risky Assets

Inputs (running case): S: E 12%, σ 20%; B: E 6%, σ 10%; ρ = 0.20 → Cov = 0.20 × 20 × 10 = **40** (%²); r_f = 3%

Concept	Formula	Keywords
Portfolio return	E(r_P) = w_S E(r_S) + w_B E(r_B)	**Linear** in weights
Covariance	Cov = Σ p_i [r_S(i) − E(r_S)][r_B(i) − E(r_B)]	Move together (+) / offset (−); units hard to read
Correlation	ρ = Cov/(σ_S σ_B); Cov = ρσ_Sσ_B	Between **−1 and +1**; unit-free
Portfolio variance	σ²_P = $w_S^2\sigma_S^2 + w_B^2\sigma_B^2 + 2\, w_S\, w_B$ Cov	Interaction term = only term that can be **negative** = only source of diversification

- ρ = +1: no benefit; ρ = −1: perfect hedge (risk-free mix possible); 0 < ρ < 1: usual case (partial benefit)
- **Cost-free**: E(r) = weighted avg (no return sacrificed) but σ_P < weighted-avg σ → benefit exists only if ρ < +1

CALC — 60/40 step order

1. E = 0.6(12) + 0.4(6) = **9.60%**
2. Stock term = 0.36 × 400 = 144.0
3. Bond term = 0.16 × 100 = 16.0

4. Interaction = 2(0.6)(0.4)(40) = 19.2
5. Variance = 179.2; σ = **13.39%** (vs weighted-avg 16.00% → saved 2.61)
6. Sharpe = (9.60 − 3)/13.39 = **0.493**

Effect of ρ (60/40; fixed terms 144 + 16)

ρ	Cov	Interaction	Variance	σ_P	Sharpe
+1.0	200	+96	256	16.00%	0.413
+0.5	100	+48	208	14.42%	0.458
+0.2	40	+19.2	179.2	13.39%	0.493
0.0	0	0	160	12.65%	0.522
−0.5	−100	−48	112	10.58%	0.624
−1.0	−200	−96	64	8.00%	0.825

- ρ = +1: **$\sigma_P = w_S\sigma_S + w_B\sigma_B$**; ρ = −1: **$\sigma_P = |w_S\sigma_S - w_B\sigma_B|$**
- **Zero-risk mix (ρ = −1)**: 20w = 10(1 − w) → **$w_S = \sigma_B/(\sigma_S + \sigma_B) = 1/3$**; E = 1/3(12) + 2/3(6) = **8%**

Scenario inputs (Example 3) — p = 0.25 each: stock −22 / +6 / +18 / +38 → E = **10%**; bond +12 / +4 / +3 / +5 → E = **6%**; deviation products −192, +8, −24, −28 → **Cov = −59**

- Steps: (1) E = Σp·r; (2) variances Σp·dev²; (3) Cov = Σp·dev_S·dev_B, ρ = Cov/(σ_Sσ_B)
- 50/50 check: σ_P = **9.57%** (formula = state-by-state); weighted avg 0.5(21.73) + 0.5(3.54) = 12.64% → gap **3.07 pts** = diversification benefit; E = 8.00%

Opportunity set & efficiency

- Curve of portfolios over w; lower ρ → bows further **left**; ρ = +1 straight line; ρ = −1 touches vertical axis
- **Efficient** = upward-sloping part from **minimum-variance portfolio**; points below = **dominated** (same σ, lower E)

Minimum-variance portfolio

- **$w_{S,min} = [\sigma_B^2 - Cov]/[\sigma_S^2 + \sigma_B^2 - 2Cov]$**
- Ex: (100 − 40)/(400 + 100 − 80) = 60/420 = **14.29%** stocks; E = **6.86%**; σ = **9.56%** (< bond's 10%)
- Adding riskier asset reduces risk if **$\rho < \sigma_B/\sigma_S$** = 10/20 = **0.50** (negative ρ not required)
- **TRAP** Min-variance ignores expected return → not the best portfolio

6.3 Optimal Risky Portfolio with a Risk-Free Asset

DEF

- Complete portfolio: **$E(r_C) = r_f + y[E(r_P) - r_f]$**; **$\sigma_C = y\sigma_P$**
- **CAL**: E(r_C) = r_f + S_P·σ_C; **slope = Sharpe ratio**
- Each risky portfolio has its own CAL; **steeper = strictly better** (more E at every σ) → maximize Sharpe ratio
- **Optimal risky portfolio P*** = **tangency** point of CAL with opportunity set; highest Sharpe; lies on efficient frontier

Candidate (r_f = 3%)	w_S	E	σ	Sharpe
All bonds	0%	6.00%	10.00%	0.300
Min-variance	14.29%	6.86%	9.56%	0.403
60/40	60%	9.60%	13.39%	0.493
All stocks	100%	12.00%	20.00%	0.450
Optimal P*	**48.15%**	**8.89%**	**11.82%**	**0.498**

CALC — optimal weights (excess returns R_S = E(r_S) − r_f, R_B)

- **$w_S^* = [R_S\sigma_B^2 - R_B Cov] / [R_S\sigma_B^2 + R_B\sigma_S^2 - (R_S + R_B)Cov]$**; $w_B^* = 1 - w_S^*$
- Ex (R_S = 9, R_B = 3): numerator 9(100) − 3(40) = **780**; denominator 900 + 1,200 − 480 = **1,620**; w_S^* = **0.4815**, w_B^* = 0.5185
- E(r_P) = 0.4815(12) + 0.5185(6) = **8.89%**; variance = 92.73 + 26.89 + 19.97 = 139.59; σ = **11.82%**; Sharpe = 5.89/11.82 = **0.498**
- Reading: higher premium ↑ weight; higher own variance ↓; high Cov ↓ diversification value; if $w^* \notin [0, 1]$ → short sale needed

CALC — complete portfolio (A = 6, P*: 8.89%, 11.82%; **decimals**)

1. **$y^* = [E(r_P) - r_f]/(A\sigma_P^2)$** = 0.05889/(6 × 0.013959) = **0.7031**
2. T-bills = 1 − y* = **29.69%**
3. Look-through stocks = 0.7031 × 0.4815 = **33.86%**; bonds = 0.7031 × 0.5185 = **36.45%** (sum 100%)
4. E(r_C) = 3 + 0.7031(5.89) = **7.14%**; σ_C = 0.7031 × 11.82 = **8.31%**

5. U = 0.0714 − ½(6)(0.0831)² = **5.07%** (vs 3% T-bills)

- A = 3 → y* = **1.41** → borrow at r_f (leveraged P*)

Four-step recipe

1. Inputs: E(r), σ, Cov / ρ
2. Trace E(r_P), σ_P over weights; efficient (upper) branch
3. Maximize Sharpe → P*
4. y* from A; split P* vs r_f

- **Only step 4 depends on investor**; steps 1–3 identical for all = **separation property**

6.4 Many Risky Assets

Efficient frontier (Markowitz)

- Highest E(r) for each σ; found by **minimizing variance for each target E(r)**
- Only branch **above global minimum-variance portfolio** is efficient
- Individual assets lie **inside** the frontier → all dominated
- **P*** = unique Sharpe-maximizing tangency portfolio; same math as 2-asset, more bookkeeping

Separation property

- (1) Determine optimal risky portfolio → technical, same for all clients
- (2) Decide how much in it (y) → personal, from risk aversion
- Implication: one research team serves many clients; questionnaire sets y not composition; target-date funds glide y down; bridge to CAPM (everyone holds P* → market portfolio)
- **Breaks down**: different inputs (disagree), borrowing rate > lending rate (kinked CAL), taxes, human capital, constraints (no short sale, mandates)

CALC — input list

- n returns + n variances + n(n − 1)/2 covariances = **n(n + 3)/2**
- n = 10 → 65; n = 50 → 1,325; n = 500 → **125,750**; n = 800 → 321,200; n = 3,000 → 4,504,500
- Problems: computation (solved) + **estimation error** ("error maximization")

Home bias

- Investors overweight home country vs world share; causes: information costs, currency risk, tax / regulation, familiarity
- Foreign assets lower the **covariance term** (different industrial structure → different systematic shocks)
- Caveats: correlations risen, currency risk; benefit ≠ 0 while ρ < +1
- Covariance argument, not forecasting

Formula sheet (PPT)

- $E(r_P) = w_S E(r_S) + w_B E(r_B)$
- $\sigma^2_P = w_S^2\sigma_S^2 + w_B^2\sigma_B^2 + 2 w_S w_B Cov$
- $Cov = \rho\sigma_S\sigma_B$
- $w_{S,min} = (\sigma_B^2 - Cov)/(\sigma_S^2 + \sigma_B^2 - 2Cov)$
- $S = (E(r_P) - r_f)/\sigma_P$
- $w_S^* = (R_S\sigma_B^2 - R_B Cov)/(R_S\sigma_B^2 + R_B\sigma_S^2 - (R_S + R_B)Cov)$
- CAL: $E(r_C) = r_f + S\cdot\sigma_C$
- $y^* = (E(r_P) - r_f)/(A\sigma_P^2)$
- **Units warning**: y* must use decimals

Out of exam scope: §6.5 single-index model, §6.6 time diversification.

Final Check: Confusable Pairs

Pair	Distinction
Primary vs secondary market	Issuer gets proceeds vs no new money
Firm commitment vs best efforts	Underwriter bears risk vs agent only
Bid vs ask	Sell at bid, buy at ask; spread = round-trip cost
Market vs limit order	Certain fill / uncertain price vs certain price / uncertain fill
Stop-loss vs stop-limit	Becomes market (price not guaranteed) vs becomes limit (fill not guaranteed)

Pair	Distinction
Long vs short margin call	Price **falls**, P = Loan/[N(1 − m)] vs price **rises**, P = C/[N(1 + m)]
BDY vs BEY vs EAY	Face / 360 simple < price / 365 simple < price / 365 compound
Discount vs add-on	T-bill, CP, BA vs CD, repo, fed funds, Eurodollar
Price vs value vs equal weight	DJIA (price) vs S&P 500 (cap) vs average of returns
Option vs futures	Right, premium, asymmetric vs obligation, margin, linear
Open-end vs closed-end vs ETF	NAV once daily vs exchange price (premium / discount) vs exchange + AP arbitrage
Mutual fund vs hedge fund	Public, 1940 Act, daily NAV vs accredited, lock-up, 2 and 20, leverage
Expense ratio vs load vs 12b-1	Annual, in NAV vs one-time (front / CDSC) vs annual distribution fee inside expense ratio
Class A vs Class C	Load + low fee (long horizon) vs no load + high fee (short horizon)
Leveraged ETF vs covered-call ETF	Daily reset decay vs capped upside, open downside
Arithmetic vs geometric vs dollar-weighted	Forecast vs past compound / manager vs IRR / investor
APR vs EAR	Scaled rate vs compounded: (1 + APR/n)^n − 1
Exact vs approximate real return	(1 + R)/(1 + i) − 1 vs R − i (overstates by r·i)
Excess return vs risk premium	Realized r − r_f vs expected E(r) − r_f
VaR vs ES	Cut-off loss vs average loss beyond cut-off
Parametric vs historical vs Monte Carlo	Normal formula vs actual past sorted vs simulated model paths
Sharpe ratio vs utility	Slope of CAL (quality of P) vs U = E − ½Aσ² (investor-specific)
CAL vs CML	Any risky P vs broad market index as P
Min-variance vs optimal (tangency)	Lowest σ, ignores E(r) vs highest Sharpe
Systematic vs firm-specific	Not diversifiable, priced vs diversifiable, unpriced
y* vs weights inside P*	Investor-specific (A) vs same for everyone (separation)

Formula one-liners

- HPR = (P_1 − P_0 + D)/P_0 · NAV = (A − L)/shares · Buying power = 1/m
- Margin call: long P = L/[N(1 − m)]; short P = C/[N(1 + m)]
- EAR = (1 + APR/n)^n − 1 · r = (1 + R)/(1 + i) − 1
- E = Σp·r · σ^2 = Σp(r − E)² · VaR = V_0 z σ √T (z: 1.645 / 2.326)
- Sharpe = (E − r_f)/σ · U = E − ½Aσ^2 · y* = (E − r_f)/(Aσ^2)
- σ^2_P = w_S²σ_S² + w_B²σ_B² + 2w_S w_B$\rho\sigma$_Sσ_B · w_min = (σ_B² − Cov)/(σ_S² + σ_B² − 2Cov)

Basis

- Scope: lecture PPT Ch.2–6 (to §6.4); Ch.1, LIBOR, §6.5, §6.6 excluded
- ※ Not in PPT (SA / MC bank only): best efforts, winner's curse, IPO long-run underperformance, 2010 flash crash, balanced fund as a named type, soft-dollar detail
- PPT graphics lost in text extraction (blank cells) were rebuilt from the formulas shown on the slides